# Elementary Statistics for Social Sciences II

**208272 (Part 1)**

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# Preface

This book covers the first half of 208272 Elementary Statistics for Social Sciences II: the principles of statistical inference, inference about means, proportions and variances, and analysis of variance. All examples use economic contexts. Unless a source is given, the numbers in the examples are hypothetical and are used for illustration only.

## How to use this book

Each section of Chapters 2–4 matches one class session and follows the same pattern:

1. **Learning objectives**
2. **Concept**: why we need this method
3. **Formulas and conditions**
4. **Worked example**: a full solution
5. **Your Turn**: a similar problem with space to write your own solution
6. **Reading R output** (from Section 3.2 onward)
7. **Formula summary**
8. **Exercises**
9. **Solutions** to Your Turn and the Exercises

In the PDF, the Your Turn problems have blank space for writing, and the solutions are at the end of each section. On the web, click a solution box to open it. Try the problem first.

| Session | Section |
|---|---|
| 1 | Chapter 1 Basic Principles of Statistical Inference |
| 2 | 2.1 Mean, $\sigma$ known |
| 3 | 2.2 Mean, $\sigma$ unknown |
| 4 | 2.3 Proportion |
| 5 | 2.4 Variance |
| 6 | 3.1 Two proportions |
| 7 | 3.2 P-values and R output; 3.3 Paired means |
| 8 | 3.4 Two variances |
| 9 | 3.5 Two independent means |
| 10 | 4.1 One-way ANOVA: CRD |
| 11 | 4.2 Multiple comparisons (Tukey); 4.3 RCBD |
| 12 | Summary: Choosing a Method |

## Notation for critical values

The subscript of a critical value is the **area to its left**. This is the same convention as the R functions `qnorm()`, `qt()`, `qchisq()` and `qf()`.

| Notation | Meaning | R |
|---|---|---|
| $z_{0.975}$ | the $z$ value with area 0.975 to its left | `qnorm(0.975)` |
| $t_{0.975,\,24}$ | the $t$ value with 24 df and area 0.975 to its left | `qt(0.975, 24)` |
| $\chi^2_{0.025,\,10}$ | the $\chi^2$ value with 10 df and area 0.025 to its left | `qchisq(0.025, 10)` |
| $F_{0.975,\,5,\,8}$ | the $F$ value with (5, 8) df and area 0.975 to its left | `qf(0.975, 5, 8)` |

A value calculated from the sample has the subscript "cal", for example $z_{\text{cal}}$, $t_{\text{cal}}$, $\chi^2_{\text{cal}}$ and $F_{\text{cal}}$.

## The five steps of a hypothesis test

1. **Hypotheses**: state $H_0$ and $H_1$.
2. **Level of significance**: state $\alpha$.
3. **Test statistic**: write the formula, substitute the values, and give the result.
4. **Critical value and critical region**.
5. **Decision and interpretation**.

## Writing the conclusion

**Decision using the critical region** (Chapters 1–2 and Section 3.1)

> $t_{\text{cal}}$ = … [falls / does not fall] in the critical region ($t_{\text{cal}}$ [<, >] …), so we [reject / do not reject] $H_0$.

**Decision using the p-value** (from Section 3.2 onward)

> p-value = … [<, ≥] $\alpha = 0.05$, so we [reject / do not reject] $H_0$.

**Interpretation**

> That is, … at the 0.05 level of significance.

- If we reject $H_0$, write $H_1$ as a sentence.

- If we do not reject $H_0$, write $H_0$ as a sentence.
- For a two-sided test that rejects $H_0$, also state the direction, using the sample statistic.

**Confidence interval**

> We are 95% confident that … is between … and ….

## Rounding

*To be added.*

# 1 Basic Principles of Statistical Inference

## Learning Objectives

*To be added.*

## 1.1 Sampling Distributions of $\bar{x}$ and $\hat{p}$

*To be added.*

## 1.2 Standard Error and the Central Limit Theorem

*To be added.*

## 1.3 The Idea of a Confidence Interval

*To be added.*

## 1.4 The Logic of Hypothesis Testing

*To be added.*

## 1.5 Type I and Type II Errors

*To be added.*

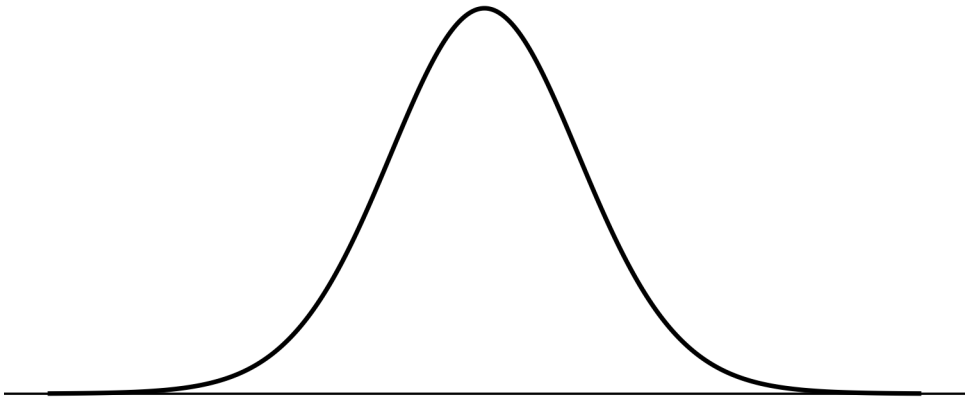


Figure 1.1: Blank curve: shade the area to the left of $t_{0.975,\,24}$ (layout demo)

## 1.6 Reading Statistical Tables

*To be added.* Shade the area to the left of the value before you read the table.

## Formula Summary

*To be added.*

## Exercises

*To be added.*

# 2 One-Sample Inference

## 2.1 Mean, $\sigma$ Known

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 2.1.1

*Problem to be added.*

**Step 1: Hypotheses**

$H_0$: _____

$H_1$: _____

**Step 2: Level of significance**

$\alpha =$ _____

**Step 3: Test statistic**

Formula: _____

Substitution: _____

Result: _____

**Step 4: Critical value and critical region**

Critical value(s): _____

Critical region: _____

**Step 5: Decision and interpretation**

Decision: _____

_____

Interpretation: _____

_____

## Formula Summary

*To be added.*

## Exercises

*To be added.*

### Solutions

**Solution: Your Turn 2.1.1**

*Solution to be added.*

## 2.2 Mean, $\sigma$ Unknown

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 2.2.1

*Problem to be added.*

**Step 1: Hypotheses**

$H_0$: _____

$H_1$: _____

**Step 2: Level of significance**

$\alpha =$ _____

**Step 3: Test statistic**

Formula: _____

Substitution: _____

Result: _____

**Step 4: Critical value and critical region**

Critical value(s): _____

Critical region: _____

**Step 5: Decision and interpretation**

Decision: _____

_____

Interpretation: _____

_____

## Formula Summary

*To be added.*

## Exercises

*To be added.*

### Solutions

**Solution: Your Turn 2.2.1**

*Solution to be added.*

## 2.3 Proportion

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 2.3.1

*Problem to be added.*

**Step 1: Hypotheses**

---

---

**Step 2: Level of significance**

**Step 3: Test statistic**

**Step 4: Critical value and critical region**

**Step 5: Decision and interpretation**

### Formula Summary

*To be added.*

### Exercises

*To be added.*

### Solutions

**Solution: Your Turn 2.3.1**

*Solution to be added.*

## 2.4 Variance

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

## Your Turn

Your Turn 2.4.1

*Problem to be added.*

**Step 1: Hypotheses**

**Step 2: Level of significance**

**Step 3: Test statistic**

**Step 4: Critical value and critical region**

**Step 5: Decision and interpretation**

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 2.4.1**

*Solution to be added.*

# 3 Two-Sample Inference

## 3.1 Two Proportions

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 3.1.1

*Problem to be added.*

**Step 1: Hypotheses**

---

---

**Step 2: Level of significance**

**Step 3: Test statistic**

**Step 4: Critical value and critical region**

**Step 5: Decision and interpretation**

### Formula Summary

*To be added.*

### Exercises

*To be added.*

### Solutions

**Solution: Your Turn 3.1.1**

*Solution to be added.*

## 3.2 P-values and R Output

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

## Your Turn

Your Turn 3.2.1

*Problem to be added.*

## Reading R Output

*To be added.*

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 3.2.1**

*Solution to be added.*

## 3.3 Paired Means

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 3.3.1

*Problem to be added.*

## Reading R Output

*To be added.*

### Formula Summary

*To be added.*

### Exercises

*To be added.*

### Solutions

**Solution: Your Turn 3.3.1**

*Solution to be added.*

## 3.4 Two Variances

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

## Your Turn

Your Turn 3.4.1

*Problem to be added.*

## Reading R Output

*To be added.*

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 3.4.1**

*Solution to be added.*

## 3.5 Two Independent Means

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

**Your Turn 3.5.1**

*Problem to be added.*

## Reading R Output

*To be added.*

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 3.5.1**

*Solution to be added.*

# 4 Analysis of Variance

## 4.1 One-Way ANOVA: CRD

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 4.1.1

*Problem to be added.*

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### Reading R Output

*To be added.*

### Formula Summary

*To be added.*

### Exercises

*To be added.*

### Solutions

**Solution: Your Turn 4.1.1**

*Solution to be added.*

## 4.2 Multiple Comparisons (Tukey)

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

## Worked Example

*To be added.*

## Your Turn

Your Turn 4.2.1

*Problem to be added.*

## Reading R Output

*To be added.*

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 4.2.1**

*Solution to be added.*

## 4.3 Randomized Complete Block Design (RCBD)

### Learning Objectives

*To be added.*

### Concept

*To be added.*

### Formulas and Conditions

*To be added.*

### Worked Example

*To be added.*

### Your Turn

Your Turn 4.3.1

*Problem to be added.*

## Reading R Output

*To be added.*

## Formula Summary

*To be added.*

## Exercises

*To be added.*

## Solutions

**Solution: Your Turn 4.3.1**

*Solution to be added.*

# Summary: Choosing a Method

## Which Method? A Flowchart

*To be added.*

## Mixed Problems

*To be added.*

## Writing a Full Conclusion

*To be added.*

# A Statistical Tables

All tables give the value with the stated **area to its left**.

## A.1 Standard Normal Distribution

*To be added.*

## A.2 $t$ Distribution

*To be added.*

## A.3 Chi-Square Distribution

*To be added.*

## A.4 $F$ Distribution

*To be added.*